



§1 Translation, Scaling and Reflection

Key Questions:

- ☐ What is a *geometrical transformation*? In particular, what is a
 - ☐ *translation*? ☐ *scaling*? ☐ *reflection*?
- ☐ What is the relationship between the graphs of $y = f(x)$ and
 - ☐ $y = f(x) + a$? ☐ $y = f(x + a)$?
 - ☐ $y = af(x)$? ☐ $y = f(ax)$?
 - ☐ $y = -f(x)$? ☐ $y = f(-x)$?
- ☐ Given the graph of $y = f(x)$, how do we sketch the graphs of the above equations (based on the description of each transformation)?

What is a *geometrical transformation*?

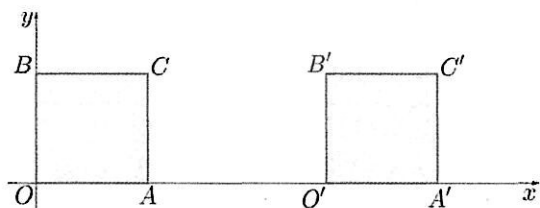
Definition 1.1 (Geometrical Transformations)

A *geometrical transformation* of an object or shape in the 2D space $\mathbb{R}^2$ is a mapping of every point on the object (or shape) to another point (not necessarily a different one) in $\mathbb{R}^2$.

What is a *translation*?

Definition 1.2 (Translation)

A *translation* in $\mathbb{R}^2$ is a geometrical transformation that moves every point of an object in $\mathbb{R}^2$ by the **same distance** in the **same direction**.

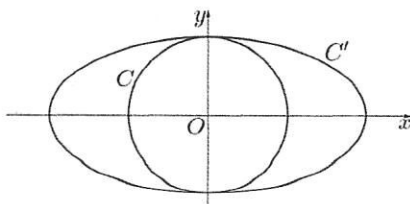


The square $O'A'C'B'$ is a translation of the square $OACB$ in the positive x -direction.

What is a *scaling*?

Definition 1.3 (Scaling)

A *scaling* in $\mathbb{R}^2$ is a geometrical transformation that increases or decreases the distance of every point on an object from a fixed line (called an *invariant line*) by a **constant multiple** (called a *scale factor*).

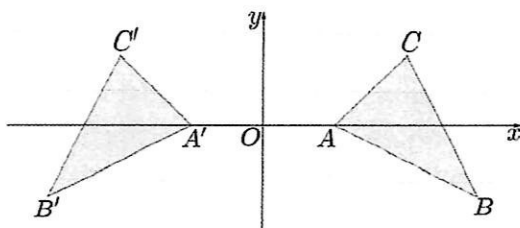


The ellipse C' is a scaling of the circle C parallel to the x -axis, with y -axis invariant.

What is a reflection?

Definition 1.4 (Reflection)

A *reflection* in $\mathbb{R}^2$ is a geometrical transformation that maps every point on an object to its mirror image in a given fixed line (or axis).



The triangle $A'B'C'$ is a reflection of the triangle ABC in the y -axis.

How is the graph of $y = f(x)$ related to the graphs of $y = f(x) + a$ and $y = f(x) - a$

Result 1.5 (Vertical Translation)

If a is a positive constant, then

- $y = f(x) + a$ is a translation of $y = f(x)$ in the **positive y -direction** by a units.
- $y = f(x) - a$ is a translation of $y = f(x)$ in the **negative y -direction** by a units.

Note the difference between the descriptions and the variable replacements.

E.g. “**positive y -direction** by a units” corresponds to “replace y with $y - a$ ”.

Note: When we replace the variable y in the equation $y = f(x)$ with the variable $y - a$, we get $y = f(x) + a$, since

$$y - a = f(x) \Rightarrow y = f(x) + a.$$

Similarly, $y = f(x) - a$ is obtained when we replace the variable y in the equation $y = f(x)$ with the variable $y + a$.

How is the graph of $y = f(x)$ related to the graphs of $y = f(x - a)$ and $y = f(x + a)$

Result 1.6 (Horizontal Translation)

If a is a positive constant, then the graph of

- $y = f(x - a)$ is a translation of $y = f(x)$ in the **positive x -direction** by a units.
- $y = f(x + a)$ is a translation of $y = f(x)$ in the **negative x -direction** by a units.

Note the difference between the descriptions and the variable replacements.

E.g. “**positive x -direction** by a units” corresponds to “replace x with $x - a$ ”, which is consistent with what we observe in Result 1.5.

Note: When we replace the variable x in the equation $y = f(x)$ with the variable $x - a$, we get $y = f(x - a)$.

Similarly, $y = f(x + a)$ is obtained when we replace the variable x in the equation $y = f(x)$ with the variable $x + a$.

How are the graphs of
 $y = f(x)$ and
 $y = af(x)$ related?

Result 1.7 (Vertical Scaling)

If a is a positive constant, then $y = af(x)$ is a scaling of
 $y = f(x)$ **parallel to the y -axis by a factor of a .**

Note the difference
 between the
 descriptions and the
 variable
 replacements.

E.g. “scaling parallel
 to y -axis by **factor of**
 a ” corresponds to

“replace y with $\frac{y}{a}$ ”.

Note: When we replace the variable y in the equation

$y = f(x)$ with the variable $\frac{y}{a}$, we get $y = af(x)$, since

$$\frac{y}{a} = f(x) \Rightarrow y = af(x).$$

How are the graphs of
 $y = f(x)$ and
 $y = f(ax)$ related?

Result 1.8 (Horizontal Scaling)

If a is a positive constant, then $y = f(ax)$ is a scaling of
 $y = f(x)$ **parallel to the x -axis by a factor of $\frac{1}{a}$.**

Note the difference
 between the
 descriptions and the
 variable
 replacements.

E.g. “scaling parallel
 to x -axis by **factor of**
 $\frac{1}{a}$ ” corresponds to

“replace x with ax ”,
 which is consistent
 with that we observe in
 Result 1.7.

Note: When we replace the variable x in the equation
 $y = f(x)$ with the variable ax , we get $y = f(ax)$.

How are the graphs of
 $y = f(x)$ and
 $y = -f(x)$ related?

Result 1.9 (Vertical Reflection)

$y = -f(x)$ is a reflection of $y = f(x)$ in the **x -axis**.

Note the difference
 between the
 descriptions and the
 variable
 replacements.

E.g. “**reflection in the**
 x -axis” corresponds
 to “replace y with $-y$ ”.

Note: When we replace the variable y in the equation
 $y = f(x)$ with the variable $-y$, we get $y = -f(x)$, since

$$-y = f(x) \Rightarrow y = -f(x).$$

How are the graphs of $y = f(x)$ and $y = f(-x)$ related?

Note the difference between the descriptions and the variable replacements.

E.g. "reflection in the y -axis" corresponds to "replace x with $-x$ ".

Result 1.10 (Horizontal Reflection)

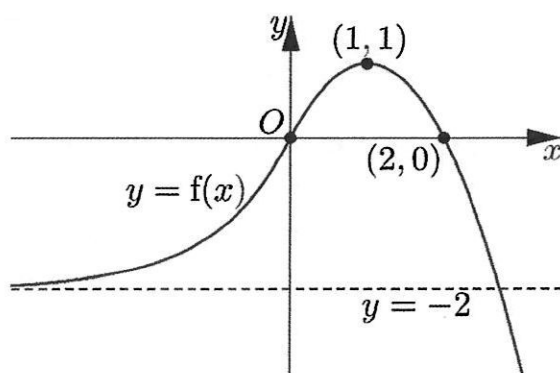
$y = f(-x)$ is a reflection of $y = f(x)$ in the y -axis.

Note: When we replace the variable x in the equation $y = f(x)$ with the variable $-x$, we get $y = f(-x)$.

Given the graph of **Exercise 1.1**

$y = f(x)$, how do we sketch the graphs of

- $y = f(x) + a$?
- $y = f(x - a)$?
- $y = af(x)$?
- $y = f(ax)$?
- $y = -f(x)$?
- $y = f(-x)$?



Based on the equations given, we can easily deduce the variable replacements and that helps us describe the corresponding transformations and, in turn, sketch the resultant graphs based on the descriptions.

We will do the same for more complex cases in Section 2.

The diagram shows the curve $y = f(x)$. The curve crosses the x -axis at the origin O and the point $(2, 0)$. It has a maximum turning point at $(1, 1)$ and an asymptote with the equation $y = -2$.

On separate diagrams, sketch the curves with equations

- | | |
|-----------------------------|----------------------|
| (a) $y = f(x) + 2$, | (b) $y = f(x + 2)$, |
| (c) $y = \frac{1}{2}f(x)$, | (d) $y = f(2x)$, |
| (e) $y = -f(x)$, | (f) $y = f(-x)$. |

For each part, label the equations of any asymptotes and the coordinates of any turning points and of the points where the curve crosses the axes.

Solution

(a)

Var Replacement:

Replace y with $y - 2$

Description:

Translate the graph of $y = f(x)$ in the positive y -direction by 2 units.

Var Replacement: (b)
Replace x with $x + 2$

Description:
Translate the graph of
 $y = f(x)$ in the
negative x -direction
by **2 units**.

Var Replacement: (c)
Replace y with $2y$

Description:
Scale the graph of
 $y = f(x)$ **parallel to**
the y -axis by a factor
of $\frac{1}{2}$.

Var Replacement: (d)
Replace x with $2x$

Description:
Scale the graph of
 $y = f(x)$ **parallel to**
the x -axis by a factor
of $\frac{1}{2}$.

Var Replacement: (e)
Replace y with $-y$

Description:
Reflect the graph of
 $y = f(x)$ **in the x -axis**.

Var Replacement: (f)
Replace x with $-x$

Description:
Reflect the graph of
 $y = f(x)$ **in the y -axis**.

§2 Composite Transformations

Key Questions:

- ☐ What is the relationship between the graphs of $y = f(x)$ and $y = af(x) + b$? ☐ $y = f(ax - b)$?
- ☐ Given the graph of $y = f(x)$, how do we sketch the graph of $y = af(cx - d) + b$ (based on the descriptions of a sequence transformations)?
- ☐ How do we formulate a sequence of transformations that transforms one given curve on to another given curve?
- ☐ How do we determine the equation of the resultant curve after a sequence of transformations is performed on a given curve?

How are the graphs of $y = f(x)$ and $y = af(x) + b$ related?

Result 2.1 (Relationship between $y = f(x)$ and $y = af(x) + b$)

If a and b are positive constants, then the graph of $y = af(x) + b$ can be obtained from the graph of $y = f(x)$ through the following sequence of transformations:

1. Scale the original graph parallel to the y -axis by a factor of a , **then**
2. translate in the positive y -direction by b units.

How do we tell that the transformations must be done in that order?

By considering the variable replacements that lead to the resultant equation (illustrated below), we will be able to deduce the sequence of transformations.

$$y = f(x)$$

↓ replace y with $\frac{y}{a}$ (corresponds to 1.)

$$\frac{y}{a} = f(x) \Rightarrow y = af(x)$$

↓ replace y with $y - b$ (corresponds to 2.)

$$y - b = af(x) \Rightarrow y = af(x) + b$$

How are the graphs of
 $y = f(x)$ and
 $y = f(ax - b)$ related?

Result 2.2 (Relationship between $y = f(x)$ and $y = f(ax - b)$)

If a and b are positive constants, then the graph of $y = f(ax - b)$ can be obtained from the graph of $y = f(x)$ through the following sequence of transformations:

1. Translate the original graph in the positive x -direction by b units, **then**
2. Scale parallel to the x -axis by a factor of $\frac{1}{a}$.

Note that when we do variable replacement, we are **literally replacing ONLY x** (or y).

How do we tell that the transformations must be done in that order?

By considering the variable replacements that lead to the resultant equation (illustrated below), we will be able to deduce the sequence of transformations.

For example, in the 2nd variable replacement of x with ax , we yield $f(ax - b)$ instead of $f(a(x - b))$.

$$y = f(x)$$

↓ replace x with $x - b$ (corresponds to 1.)

$$y = f(x - b)$$

↓ replace x with ax (corresponds to 2.)

$$y = f(ax - b)$$

Questions to Ponder

1. If the order of transformations in Results 2.1 and 2.2 is reversed, would the same graphs still be obtained? Why or why not?
2. If a and/or b are/is negative, how would Results 2.1 and 2.2 change?
3. To obtain the graph of $y = f(x - a) + b$ from the graph of $y = f(x)$, does the order of the horizontal and vertical translations matter? Why or why not?

To obtain the graph of $y = bf(ax)$ from the graph of $y = f(x)$, does the order of the horizontal and vertical scalings matter? Why or why not?

To obtain the graph of $y = -f(-x)$ from the graph of $y = f(x)$, does the order of the horizontal and vertical reflection matter? Why or why not?

Answers

1. No, a different graph will be obtained. For example, if the order is reversed in Result 2.2, then the equation of the resultant graph would be

$$y = f(x)$$

↓ replace x with ax

$$y = f(ax)$$

↓ replace x with $x - b$

$$y = f(a(x - b)) = f(ax - ab)$$

If we were to stick to this reversed order of scaling followed by translation, the correct graph of $y = f(ax - b)$ can be obtained in this case by changing the translation distance to $\frac{b}{a}$ units.

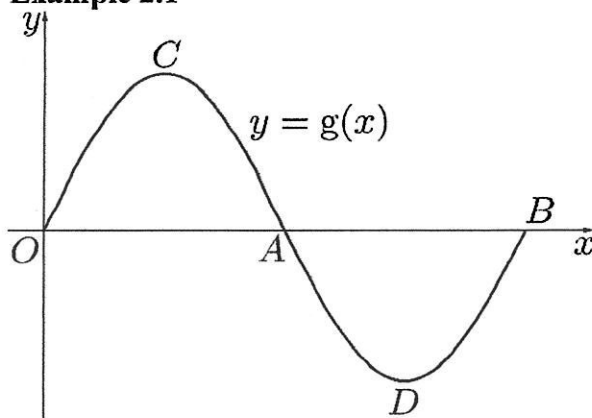
2. If b is negative, then the translations in Results 2.1 and 2.2 will be in the negative y - and negative x -directions respectively.

If a is negative, then the scaling in Result 2.1 will be replaced with a scaling by a factor of $-a$ parallel to the y -axis, followed by a reflection in the x -axis. The scaling in Result 2.2 will be replaced with a scaling by a factor of $-\frac{1}{a}$ parallel to the x -axis, followed by a reflection in the y -axis.

Note that $-a$ and $-\frac{1}{a}$ are positive constants, since a is negative.

3. No, the order does not matter between vertical and horizontal transformations. Vertical transformations affect only the y -coordinate of each point on the curve while horizontal transformations affect only the x -coordinate of each point on the curve. Thus, we can switch the order between vertical and horizontal transformations without changing the resultant graph.

Given the graph of **Example 2.1**
 $y = f(x)$, how do we
 sketch the graph of
 $y = af(cx-d)+b$?



The diagram shows the curve $y = g(x)$. The curve crosses the x -axis at the points A and B , and has a maximum and minimum turning point at C and D respectively. The coordinates of A , B , C and D are $(\pi, 0)$, $(2\pi, 0)$, $(\frac{\pi}{2}, 1)$ and $(\frac{3\pi}{2}, -1)$ respectively.

On separate diagrams, sketch the curves with equations

(a) $y = \frac{1}{2}g(x) - \frac{1}{2}$.

(b) $y = g(\pi - 2x)$,

(c) $y = 2g\left(x - \frac{\pi}{3}\right)$.

For each part, label the coordinates of the points corresponding to each of O , A , B , C and D after the transformation.

Suggest a possible equation for the curve in part (c) in the form of

$$y = a \sin x + b \cos x,$$

where a and b are real constants to be determined exactly.

Solution

As with Example 1.1, we first determine the variable replacements that will algebraically transform the original equation to the resultant equation.

Then based on these variable replacements, we can geometrically describe the transformation (in that order) which helps us with the sketching of the resultant curves.

In this example (for (a) and (b) only), we show the resultant curve after each transformation for the purpose of visualisation. You are not required to do so in test/examination if you are able to give the final graph directly.

- (a) To obtain $y = \frac{1}{2}g(x) - \frac{1}{2}$ from $y = g(x)$, we do the following:

$$y = g(x)$$

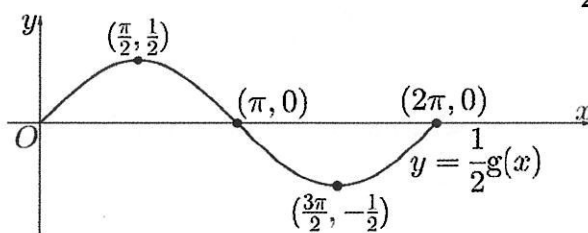
↓ replace y with $2y$ --- (I)

$$2y = g(x) \Rightarrow y = \frac{1}{2}g(x)$$

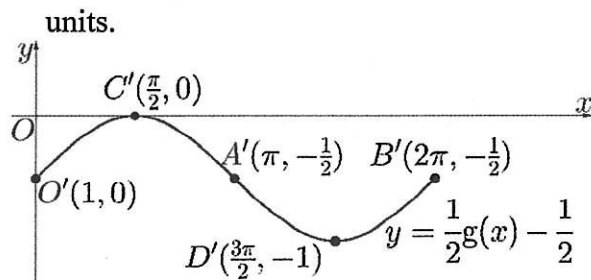
↓ replace y with $y + \frac{1}{2}$ --- (II)

$$y + \frac{1}{2} = \frac{1}{2}g(x) \Rightarrow y = \frac{1}{2}g(x) - \frac{1}{2}$$

- (I) Scale parallel to the y -axis by a factor of $\frac{1}{2}$.



- (II) Translate in the negative y -direction by $\frac{1}{2}$ units.



Track the coordinates of the transformed points based on the descriptions of the transformations to check the accuracy of your sketch. Workings for the check are demonstrated in (a) and (b) only. You are not required to do so in test/examination if you are confident of your final graph.

Check: The coordinates of the points corresponding to A , B , C and D are as follows:

$$O(0, 0) \xrightarrow{\text{I}} (0, 0) \xrightarrow{\text{II}} O'\left(0, -\frac{1}{2}\right)$$

$$A(\pi, 0) \xrightarrow{\text{I}} (\pi, 0) \xrightarrow{\text{II}} A'\left(\pi, -\frac{1}{2}\right)$$

$$B(2\pi, 0) \xrightarrow{\text{I}} (2\pi, 0) \xrightarrow{\text{II}} B'\left(2\pi, -\frac{1}{2}\right)$$

$$C\left(\frac{\pi}{2}, 1\right) \xrightarrow{\text{I}} \left(\frac{\pi}{2}, \frac{1}{2}\right) \xrightarrow{\text{II}} C'\left(\frac{\pi}{2}, 0\right)$$

$$D\left(\frac{3\pi}{2}, -1\right) \xrightarrow{\text{I}} \left(\frac{3\pi}{2}, -\frac{1}{2}\right) \xrightarrow{\text{II}} D'\left(\frac{3\pi}{2}, -1\right)$$

- (b) To obtain $y = g(\pi - 2x)$ from $y = g(x)$, we do the following:

$$y = g(x)$$

↓ replace x with $x + \pi$ --- (I)

$$y = g(x + \pi)$$

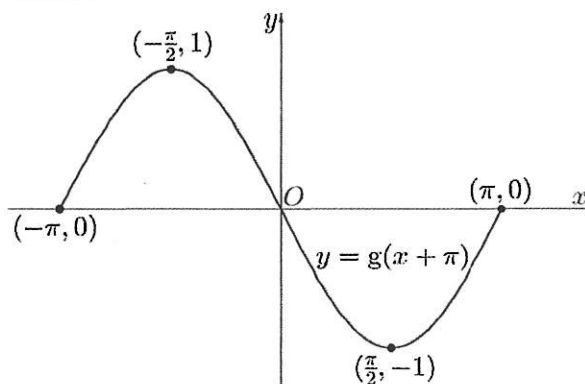
↓ replace x with $2x$ --- (II)

$$y = g(2x + \pi)$$

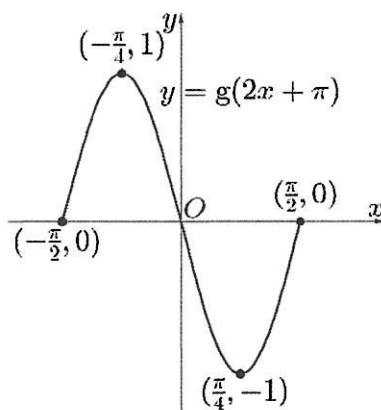
↓ replace x with $-x$ --- (III)

$$y = g(\pi - 2x)$$

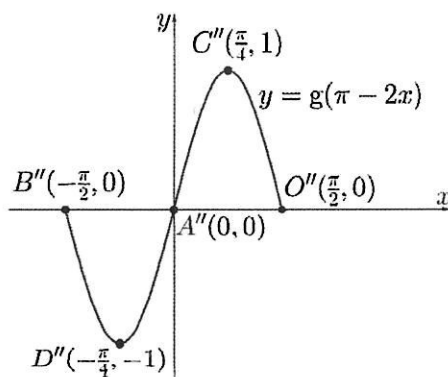
- (I) Translate in the negative x -direction by π units.



- (II) Scale parallel to the x -axis by a factor of $\frac{1}{2}$.



- (III) Reflect in the y -axis.



Check: The coordinates of the points corresponding to A , B , C and D are as follows:

$$O(0,0) \xrightarrow{\text{I}} (-\pi, 0) \xrightarrow{\text{II}} \left(-\frac{\pi}{2}, 0\right) \xrightarrow{\text{III}} O''\left(\frac{\pi}{2}, 0\right)$$

$$A(\pi, 0) \xrightarrow{\text{I}} (0, 0) \xrightarrow{\text{II}} (0, 0) \xrightarrow{\text{III}} A''(0, 0)$$

$$B(2\pi, 0) \xrightarrow{\text{I}} (\pi, 0) \xrightarrow{\text{II}} \left(\frac{\pi}{2}, 0\right) \xrightarrow{\text{III}} B''\left(-\frac{\pi}{2}, 0\right)$$

$$C\left(\frac{\pi}{2}, 1\right) \xrightarrow{\text{I}} \left(-\frac{\pi}{2}, 1\right) \xrightarrow{\text{II}} \left(-\frac{\pi}{4}, 1\right) \xrightarrow{\text{III}} C''\left(\frac{\pi}{4}, 1\right)$$

$$D\left(\frac{3\pi}{2}, -1\right) \xrightarrow{\text{I}} \left(\frac{\pi}{2}, -1\right) \xrightarrow{\text{II}} \left(\frac{\pi}{4}, -1\right) \xrightarrow{\text{III}} D''\left(-\frac{\pi}{4}, -1\right)$$

(c) To obtain $y = 2g\left(x - \frac{\pi}{3}\right)$ from $y = g(x)$, we do the following:

$$y = g(x)$$

↓ replace y with $\frac{1}{2}y$ --- (I)

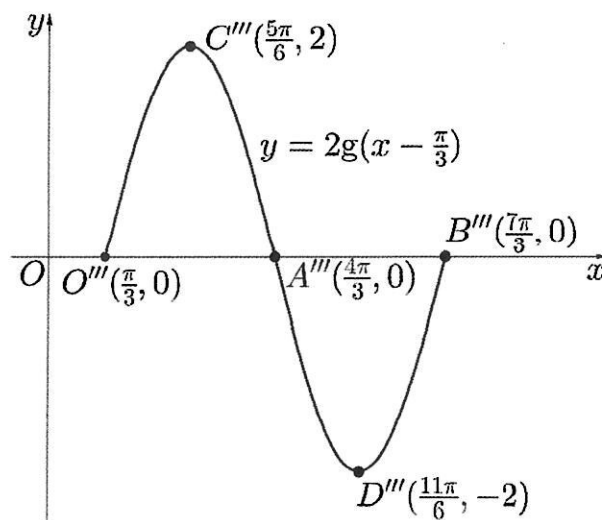
$$\frac{1}{2}y = g(x) \Rightarrow y = 2g(x)$$

↓ replace x with $x - \frac{\pi}{3}$ --- (II)

$$y = 2g\left(x - \frac{\pi}{3}\right)$$

(I) Scale parallel to the y -axis by a factor of 2.

(II) Translate in the positive x -direction by $\frac{\pi}{3}$ units.



From observation, $g(x)$ may be $\sin x$, i.e.
 $y = g(x) = \sin x$.

Thus, a possible equation for $y = 2g\left(x - \frac{\pi}{3}\right)$ is

$$\begin{aligned} y &= 2 \sin\left(x - \frac{\pi}{3}\right) \\ &= 2 \left(\sin x \cos \frac{\pi}{3} - \cos x \sin \frac{\pi}{3} \right) \\ &= 2 \left(\frac{1}{2} \sin x - \frac{\sqrt{3}}{2} \cos x \right) \\ &= \sin x - \sqrt{3} \cos x \end{aligned}$$

How do we formulate a sequence of transformations that transforms one given curve on to another given curve?

Example 2.2 (Describing a Sequence of Transformations)

Given that a , b and c are positive constants, describe a sequence of transformations that transforms the curve $y = x^2$ on to

(a) $y = ax^2 + c$,

(b) $y = ax^2 - bx$,

giving the magnitudes of your transformations in terms of a , b and/or c .

Solution

As before, we will determine the variable replacements that will algebraically transform the original equation to the resultant equation.

Then based on these variable replacements, we can geometrically describe the transformations in that order.

(a) To obtain $y = ax^2 + c$, from $y = x^2$ we do the following:

$$y = x^2$$

↓ replace y with $\frac{y}{a}$

$$\frac{y}{a} = x^2 \Rightarrow y = ax^2$$

↓ replace y with $y - c$

$$y - c = ax^2 \Rightarrow y = ax^2 + c$$

Thus, a possible sequence of transformations is:

1. Scale parallel to the y -axis by a factor of a .
2. Translate in the positive y -direction by c units.

If the replacements of variables are not immediately apparent, apply a suitable algebraic technique to express the equation in the form of $y = af(cx - d) + b$ first.

(b) By completing the square, we have

$$ax^2 - bx = a\left(x - \frac{b}{2a}\right)^2 - \frac{b^2}{4a}$$

To obtain $y = ax^2 - bx$, from $y = x^2$ we do the following:

$$y = x^2$$

↓ replace x with $x - \frac{b}{2a}$

$$y = \left(x - \frac{b}{2a}\right)^2$$

↓ replace y with $\frac{y}{a}$

$$\frac{y}{a} = \left(x - \frac{b}{2a}\right)^2 \Rightarrow y = a\left(x - \frac{b}{2a}\right)^2$$

↓ replace y with $y + \frac{b^2}{4a}$

$$y + \frac{b^2}{4a} = a\left(x - \frac{b}{2a}\right)^2 \Rightarrow y = a\left(x - \frac{b}{2a}\right)^2 - \frac{b^2}{4a}$$

Thus, a possible sequence of transformations is:

1. Translate in the positive x -direction by $\frac{b}{2a}$ units.
2. Scale parallel to the y -axis by a factor of a .
3. Translate in the negative y -direction by $\frac{b^2}{4a}$ units.

How do we determine the equation of the resultant curve after a sequence of transformations is performed on a given curve?

Example 2.3 (Determining the Equation of the Resultant Curve)

The curve $y = \ln x$ undergoes the following sequence of transformations in succession.

1. a scaling parallel to the y -axis by a factor of 3,
2. a translation of 1 unit in the positive y -direction.

Previously, we determine the variable replacements first, then describe the transformations based on that. In this example, we need to do the reverse.

Find the equation of the resultant curve in the form

$$y = \ln(kx^n)$$

for some constants k and n to be determined exactly.

Solution

Based on the descriptions given in the question, we determine the corresponding variable replacements, which are algebraically applied to obtain the equation of the resultant curve.

$$y = \ln x$$

↓ replace y with $\frac{y}{3}$

$$\frac{y}{3} = \ln x \Rightarrow y = 3 \ln x$$

↓ replace y with $y - 1$

$$y - 1 = 3 \ln x \Rightarrow y = 3 \ln x + 1$$

Since $y = 3 \ln x + 1 = \ln(x^3) + \ln e = \ln(ex^3)$,
then, $k = e$ and $n = 3$.

§3 Graphs of $y = |f(x)|$ and $y = f(|x|)$

Key Questions:

- ☐ What is the relationship between the graphs of $y = f(x)$ and
 - ☐ $y = |f(x)|$?
 - ☐ $y = f(|x|)$?
- ☐ Given the graph of $y = f(x)$, how do we sketch the graphs of $y = |f(x)|$ and $y = f(|x|)$?
- ☐ How do we carry out transformations that involve both the modulus function and translation/scaling/reflection?

Recall that the *modulus* (or *absolute value*) function is defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

Thus,

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0, \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

From the above, we can infer the following results:

How are the graphs of $y = f(x)$ and $y = |f(x)|$ related?

This is **NOT** the same as replacing y with $|y|$. So what actually happens when the variable y is replaced by $|y|$? Explore this in Tutorial PQ8.

Result 3.1 (Relationship between $y = f(x)$ and $y = |f(x)|$)

The graph of $y = |f(x)|$ can be obtained from the graph of $y = f(x)$ by

- **preserving** all parts of the graph of $y = f(x)$ that lie above the x -axis, and
- **reflecting** all other parts of the graph of $y = f(x)$ in the x -axis.

How are the graphs of
 $y = f(x)$ and
 $y = f(|x|)$ related?

Var Replacement:
 Replace x with $|x|$

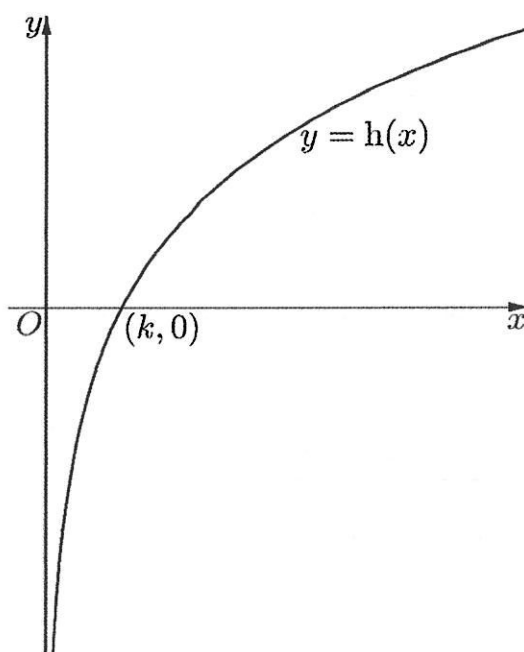
Result 3.2 (Relationship between $y = f(x)$ and $y = f(|x|)$)

The graph of $y = f(|x|)$ can be obtained from the graph of $y = f(x)$ by

- **omitting** the part of the graph of $y = f(x)$ that lies on the left of the y -axis, and
- **preserving** the part of the graph of $y = f(x)$ that lie to the right of the y -axis, and
- **reflecting** this same part in the y -axis as well.

Given the graph of
 $y = f(x)$, how do we
 sketch the graph of
 $y = |f(x)|$ and
 $y = f(|x|)$?

Example 3.1



The diagram shows the curve $y = h(x)$. The y -axis is an asymptote of the curve, which crosses the x -axis at the point $(k, 0)$.

On separate diagrams, sketch the curves with equations

(a) $y = |h(x)|$.

(b) $y = h(|x|)$.

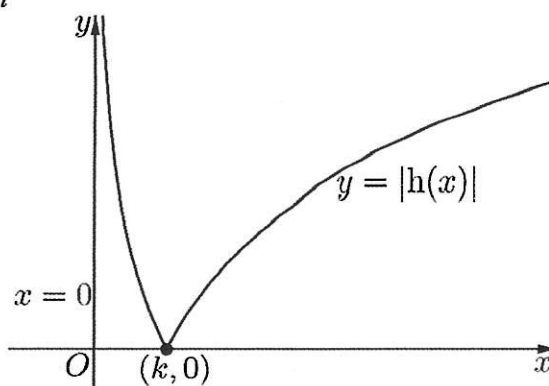
For each part, label the equations of any asymptotes and the coordinates of any turning points and of the points where the curve crosses the axes.

If given further that $h(x) = 2 \ln x$, state an equation of the curve in part (b) in non-modulus form.

*Solution***(a)**

Preserve all parts of the graph of $y = h(x)$ that lie above the x -axis.

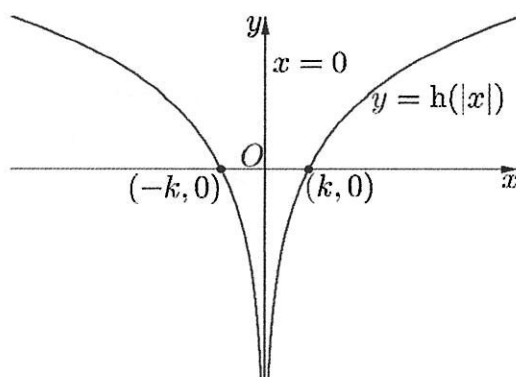
Reflect all other parts of the graph of $y = h(x)$ in the x -axis.



Omit the part of the graph of $y = h(x)$ that lies on the left of the y -axis.

Preserve the part of the graph of $y = h(x)$ that lie to the right of the y -axis.

Reflect this same part in the y -axis as well.



If $h(x) = 2 \ln x$, then an equation of the curve in part **(b)** is

$$\begin{aligned} y &= 2 \ln(|x|) \\ &= \ln(|x|^2) \\ &= \ln(x^2), \quad x \neq 0 \end{aligned}$$

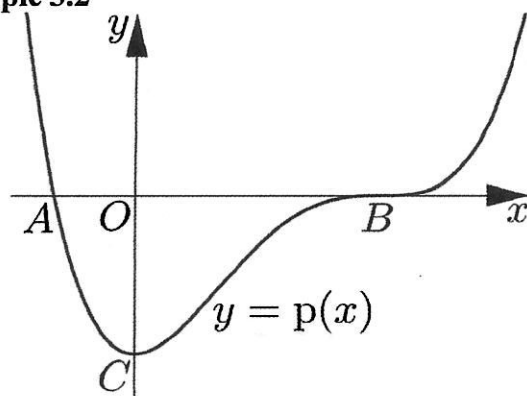
Big Idea : Transformation

Transformation refers to changes made to a mathematical object using clearly defined rule. When an object is transformed, its properties may or may not change.

Understanding the nature and effects of these transformations enables us to develop insights into the relationship between the transformed object and the original object and to use these relationships to develop methods to solve problems.

How do we carry out transformations that involve both the modulus function and translation / scaling / reflection?

Example 3.2



The diagram shows the curve $y = p(x)$. The curve crosses the x -axis at A , as well as at B , which is a stationary point of inflexion of the curve. The curve also has a minimum turning point at C where it crosses the y -axis. The coordinates of A , B and C are $(-a, 0)$, $(b, 0)$ and $(0, -c)$ respectively.

On separate diagrams, sketch the curves with equations

(a) $y = |p(x)|$.

(b) $y = p(-|x|)$.

For each part, label the coordinates of the points corresponding to each of A , B and C after the transformation.

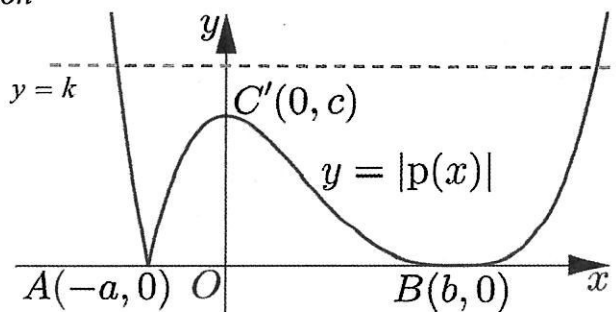
Using your sketch in part (a), determine the values of k such that the equation $|p(x)| = k$ has exactly 2 distinct solutions, giving your answer in terms of c .

Solution

Preserve all parts of the graph of $y = p(x)$ that lie above the x -axis.

Reflect all other parts of the graph of $y = p(x)$ in the x -axis.

(a)



By drawing a horizontal line $y = k$ in the same diagram in part (a) (as shown above), we see that $y = k$ has exactly 2 distinct solutions if

$$k = 0 \quad \text{or} \quad k > c.$$

As with the examples in §2, we first determine the variable replacements that will algebraically transform the original equation to the resultant equation.

Then based on these variable replacements, we can geometrically describe the transformation (in that order) which helps us with the sketching of the resultant curve.

In test/examination, it is not necessary to write out the descriptions if they are not required by the question though we need the descriptions to sketch the transformed curve.

- (b) To obtain $y = p(-|x|)$ from $y = p(x)$ we do the following:

$$y = p(x)$$

↓ replace x with $-x$

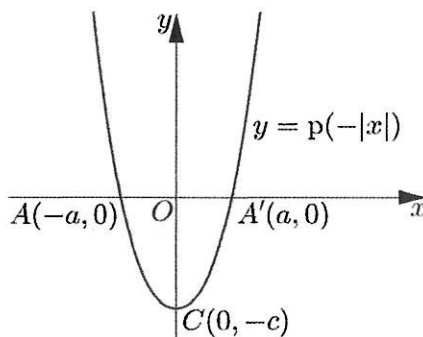
$$y = p(-x)$$

↓ replace x with $|x|$

$$y = p(-|x|)$$

Thus, a possible sequence of transformations is:

1. Reflect in the y -axis.
2. Omit the part of the graph that lies on the left of the y -axis.
3. Preserve the part of the graph that lie to the right of the y -axis.
4. Reflect this same part in the y -axis as well.



Question to Ponder

In part (a), why is point A a kink (“sharp”) point for $y = |p(x)|$ but not point B ?

Answer

The curve $y = p(x)$ is sloping downwards at point A . This implies that the gradient of the curve at A has a non-zero negative value.

Reflecting the part of the curve $y = p(x)$ below the x -axis “upwards” causes the curve $y = |p(x)|$ to have unequal gradient values **just** to the left and to the right of point A (negative and positive gradients to the left and right of point A respectively). Thus, this results in the gradient of $y = |p(x)|$ to become undefined at point A , and creates a kink point at A .

On the other hand, since point B is a stationary point of $y = p(x)$, the gradient of the curve there is **zero**. This property is preserved even after reflecting the curve $y = p(x)$ below the x -axis “upwards”. Hence the curve $y = |p(x)|$ is smooth at point B .

§4 Graph of $y = \frac{1}{f(x)}$

Key Questions:

- ☐ What is the relationship between the graphs of $y = f(x)$ and $y = \frac{1}{f(x)}$?
- ☐ Given the graph of $y = f(x)$, how do we sketch the graph of $y = \frac{1}{f(x)}$?

R/s between $y = f(x)$

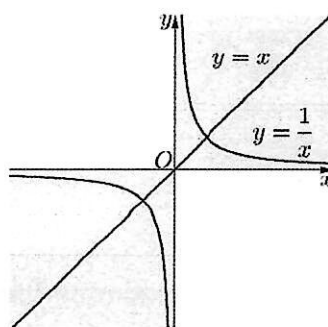
and $y = \frac{1}{f(x)}$:

How are the signs of $f(x)$ and $\frac{1}{f(x)}$ related?

Result 4.1 (Sign of $\frac{1}{f(x)}$)

1. $f(x) > 0 \Leftrightarrow \frac{1}{f(x)} > 0$
2. $f(x) < 0 \Leftrightarrow \frac{1}{f(x)} < 0$

For example, we see from the diagram here that both $y = x$ and $y = \frac{1}{x}$ lie above the x -axis for the same interval of values of x , $x > 0$.



Similarly, they both lie below the x -axis for $x < 0$.

R/s between $y = f(x)$

and $y = \frac{1}{f(x)}$:

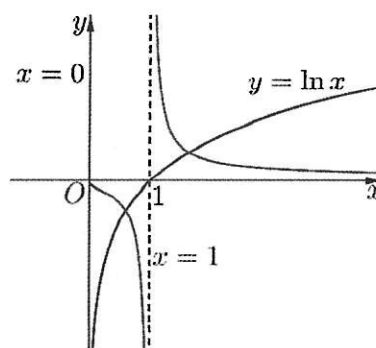
How are the x -intercepts and vertical asymptotes of $y = f(x)$ and $y = \frac{1}{f(x)}$ related?

Result 4.2 (x -Intercepts and Vertical Asymptotes of $y = \frac{1}{f(x)}$)

Between the graphs of $y = f(x)$ and $y = \frac{1}{f(x)}$,
 $(a, 0)$ is an x -intercept of one of them
 $\Updownarrow$
 $x = a$ is a vertical asymptote of the other.

Between $y = f(x)$ and $y = \frac{1}{f(x)}$, if one

approaches 0 in value, the other will approach $+\infty$ or $-\infty$. Thus, the vertical asymptotes of one curve, if any, will occur **exactly** at the x -intercepts of the other, if any.



R/s between $y = f(x)$

and $y = \frac{1}{f(x)}$:

How are the y -values of $y = f(x)$ and

$y = \frac{1}{f(x)}$ related?

Result 4.3 (y -values of $y = \frac{1}{f(x)}$)

Between the graphs of $y = f(x)$ and $y = \frac{1}{f(x)}$,

(a, b) lies on one of them

$\Updownarrow$

$\left(a, \frac{1}{b}\right)$ lies on the other,

for non-zero b .

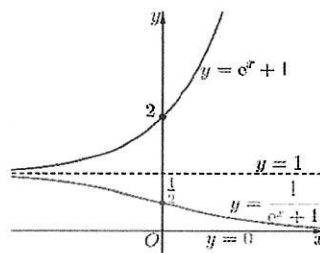
In particular, the y -intercept of

$y = \frac{1}{f(x)}$ is the reciprocal of

the y -intercept of $y = f(x)$.

Also, if $f(x) > 1$, then

$0 < \frac{1}{f(x)} < 1$ (and vice versa).



Result 4.3 can be extended to the following two results.

R/s between $y = f(x)$

and $y = \frac{1}{f(x)}$:

How are the horizontal asymptotes of $y = f(x)$ and

$y = \frac{1}{f(x)}$ related?

Result 4.4 (Horizontal Asymptotes of $y = \frac{1}{f(x)}$)

Between the graphs of $y = f(x)$ and $y = \frac{1}{f(x)}$,

$y = b$ is a horizontal asymptote of one of them

$\Updownarrow$

$y = \frac{1}{b}$ is a horizontal asymptote of the other,

for non-zero b .

What happens in the graph of $y = \frac{1}{f(x)}$ if $y = f(x)$ has a horizontal asymptote $y = 0$ (i.e. $b = 0$)?

R/s between $y = f(x)$

and $y = \frac{1}{f(x)}$:

How are the turning points of $y = f(x)$ and

$y = \frac{1}{f(x)}$ related?

Result 4.5 (Turning Points of $y = \frac{1}{f(x)}$)

Between the graphs of $y = f(x)$ and $y = \frac{1}{f(x)}$,

(a, b) is a max (or min) point of one of them

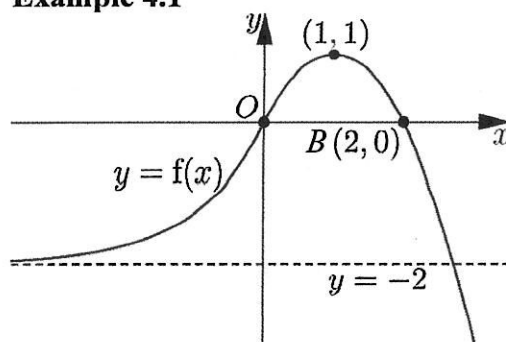
$\Updownarrow$

$\left(a, \frac{1}{b}\right)$ is a min (or max) point of the other,

for non-zero b . (Why?)

For the case where $b = 0$, refer to **Result 4.2**.

Given the graph of $y = f(x)$, how do we sketch the graph of $y = \frac{1}{f(x)}$?

Example 4.1

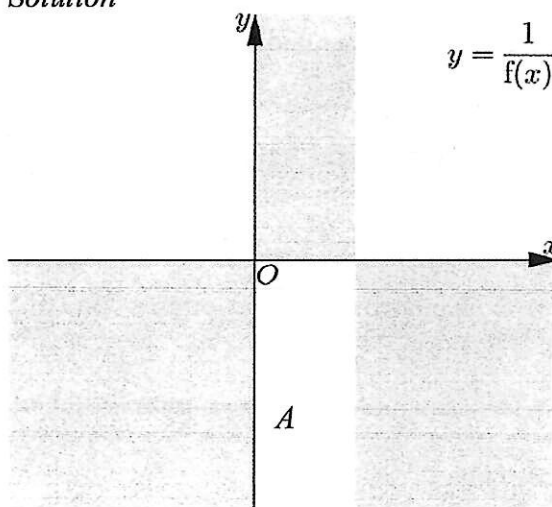
The diagram shows the curve $y = f(x)$. The curve crosses the x -axis at the origin O and the point $B(2, 0)$. It has a maximum turning point at $A(1, 1)$, and an asymptote with the equation $y = -2$. Sketch the curve $y = \frac{1}{f(x)}$.

$$y = \frac{1}{f(x)}$$

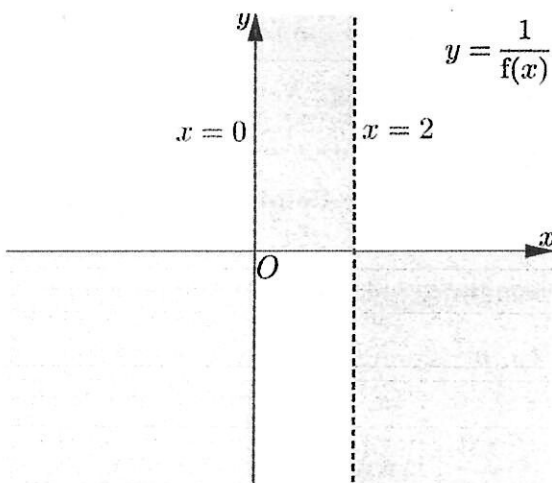
Solution

Step 1: (Result 4.1)

Using the signs of $f(x)$, demarcate imaginary boxes in your sketch for where the graph of $y = \frac{1}{f(x)}$ must lie in.

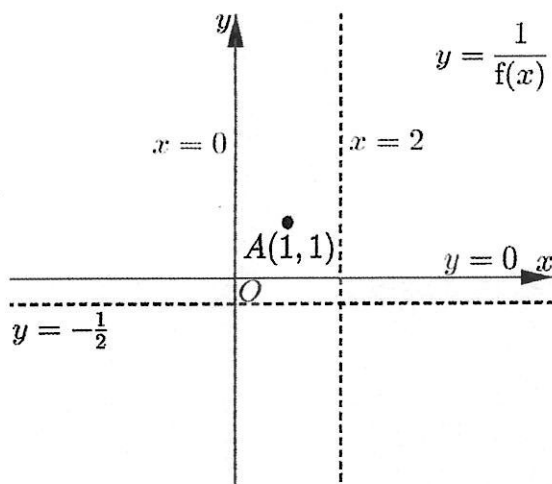
Step 2: (Result 4.2)

Mark out the x -intercepts and/or vertical asymptotes.



Step 3:*(Results 4.3 – 4.5)*

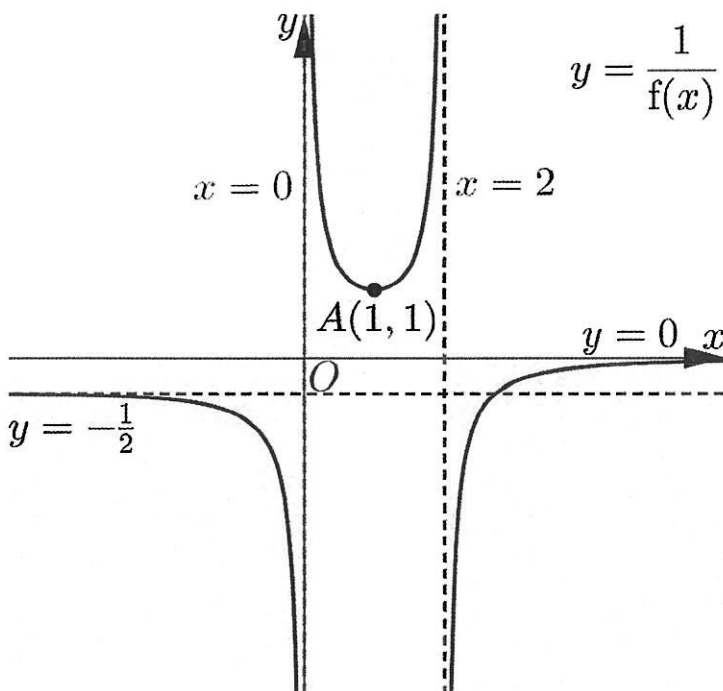
Mark out the y -intercepts, turning points and horizontal asymptotes, if any.

**Step 4:**

The imaginary boxes and asymptotes act as a skeletal structure.

Bearing in mind that

$\frac{1}{f(x)}$ and $f(x)$ are such that as one increases, the other decreases, sketch the curve by fitting it “naturally” into this structure.

**Question to Ponder**

Why does the curve $y = \frac{1}{f(x)}$ approach the x -axis for $x \rightarrow +\infty$?

Answer

In the original curve $y = f(x)$,

$$f(x) \rightarrow -\infty \text{ as } x \rightarrow +\infty.$$

Therefore,

$$\frac{1}{f(x)} \rightarrow 0 \text{ as } x \rightarrow +\infty.$$

This implies that the curve $y = \frac{1}{f(x)}$ approaches the x -axis as $x \rightarrow +\infty$.